

|                 |                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
|                 | As such, the acceleration is insignificant and thus the car's motion is not affected.                                           |
| <b>3(a)(i)</b>  | $F_E = qE = \frac{qV}{d}$ $= \frac{(3.2 \times 10^{-19})(900)}{(3.6 \times 10^{-2})}$ $= 8.0 \times 10^{-15} \text{ N (shown)}$ |
| <b>3(a)(ii)</b> | Applying Newton's 2 <sup>nd</sup> Law,                                                                                          |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | $F_E = ma$ $a = \frac{F_E}{m}$ $= \frac{8.0 \times 10^{-15}}{6.6 \times 10^{-27}}$ $= 1.2 \times 10^{12} \text{ m s}^{-2}$                                                                                                                                                                                                                                                                                                                                                                   |
| <b>3(b)</b>    | <p>For horizontal motion,</p> $s_x = u_x t$ $t = \frac{s_x}{u_x} = \frac{7.5 \times 10^{-2}}{4.1 \times 10^5}$ $t = 1.8 \times 10^{-7} \text{ s}$ <p>For vertical motion,</p> $s_y = u_y t + \frac{1}{2} a_y t^2$ $= 0 + \frac{1}{2} (1.2 \times 10^{12}) (1.8 \times 10^{-7})^2 = 0.020 \text{ m}$ <p>The vertical distance between the point of entry and the lower plate is 0.018m.<br/>As the vertical displacement of 0.020m is more than 0.018m, the particle will hit the plates.</p> |
| <b>4(a)(i)</b> | Magnetic flux density is defined as the force acting on a conductor of unit length                                                                                                                                                                                                                                                                                                                                                                                                           |